Pratyush Bhatt

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Education

The Pennsylvania State University — State College, PA
Bachelor of Science in Mechanical Engineering — *Expected Graduation May 2027*

Experience

Mechanical Engineering Intern — June 2026 – July 2026
Baxter International — *Batesville, IN*

- Designed and developed a modular experimental test apparatus for porous media characterization from concept to prototype, applied Design for Manufacturing (DFM) principles and created detailed 3D CAD models and assemblies in Solid Edge.
- Performed Finite Element Analysis to optimize structural stiffness and load distribution, while supporting CFD-based porous media analysis for Darcy–Forchheimer coefficient (A & B) determination and pressure-drop characterization.
- Redesigned the fixture's sealing system by optimizing gasket geometry and clamp configuration, reducing air leakage from 35% to less than 10%, significantly improving test accuracy and sealing performance.

Research Assistant – Locomotion in Biology and Robotics Lab — October 2025 – Present
The Pennsylvania State University — *State College, PA*

- Working on a six-legged hexapod robot to study locomotion efficiency by comparing alternating tripod gait patterns.
- Collecting and analyzing experimental data using MATLAB and Vicon Nexus across different phase shifts and duty cycles to optimize gait performance.
- Assisting in constructing and modifying hexapods, including designing mechanical adjustments, calibrating sensors, and implementing optimized gait patterns in lab testing.

Course Grader – ME 330: Engineering Design and Analysis — January 2026 – Present
The Pennsylvania State University — *State College, PA*

- Evaluating assignments of 200 students and providing feedback on SolidWorks modeling and simulation coursework.
- Assisted students in understanding CAD modeling and finite element analysis (FEA)

Shell Eco-Marathon Club — September 2025 – January 2026
The Pennsylvania State University — *State College, PA*

- Contributed to chassis and ergonomic design, including the structural frame, driver seat, and driver-controlled system integration, for an Urban Concept vehicle that achieved a team fuel-economy result of 432.32 mpg.
- Applied CAD, manufacturing, and hands-on testing to balance fuel economy, practicality, and safety for a competition-ready vehicle.

Projects

BattleBots x NAISS Robotics Competition | *Mechanical Design* — October 2025 – January 2026

- Designed and fabricated the robot chassis within a strict 6 lb weight limit, focusing on strength and durability.
- Created CAD models in Autodesk Fusion 360 to optimize structural integrity and integration with driving and weapon systems.
- Collaborated with cross-functional teams to integrate mechanical, electrical, and control systems.

Visison guided robotic Arm | *Embedded Systems* — August 2026 – October 2026

- Designed a 3-DOF robotic arm, focusing on joint geometry, mechanical alignment, and structural stability.
- Configured a ROS 2 robot model for visualization in RViz, simulation in Gazebo, and motion planning in MoveIt 2; calibrated joint limits.
- Developed an OpenCV perception pipeline to detect objects and transform image coordinates into robot coordinates; implemented inverse kinematics for autonomous pick-and-place execution.
- Validated end-to-end operation over 40 physical trials, achieving 35 successful pick-and-place cycles (87.5% success rate).

Self-Balancing Robot with PID Control | *Feedback control* — September 2026 – October 2026

- Developed a two-wheel inverted-pendulum robot integrating mechanical design and embedded programming.
- Implemented complementary-filter sensor fusion and real-time PD feedback control in C++ to estimate pitch and generate bidirectional PWM motor commands.
- Calibrated sensor offsets and tuned proportional and derivative gains; incorporated motor deadband compensation and tilt-triggered motor shutdown.

Technical Skills

CAD: SolidWorks, Solid Edge, Autodesk Fusion 360, AutoCAD
Analysis and Simulation: ANSYS, MATLAB, Vicon Nexus
Robotics and Embedded Systems: ROS 2, MoveIt 2, OpenCV, Gazebo, RViz
Programming: Python, MATLAB, C++